Waiting for the first success

Upper high school · probability and sequences · Worked solutions · 08

Exercise 1. With $p = \frac{1}{4}$, find $P(T = 3)$.

Solution. $(\frac{3}{4})^2(\frac{1}{4}) = \frac{9}{64}$.

Exercise 2. With $p = \frac{1}{4}$, find $P(T \leq 3)$.

Solution. $1 - P(T > 3) = 1 - (\frac{3}{4})^3 = \frac{37}{64}$.

Exercise 3. For $0 < p < 1$, find $P(T > m + n | T > m)$.

Solution. The ratio of tail probabilities is $\frac{(1-p)^{m+n}\, m}{1-p} = (1-p)^n$. After m failures, the remaining waiting-time distribution is unchanged.

Exercise 4. Success has probability 0.2 at each independent trial. What is the mean waiting time? Must the fifth trial succeed?

Solution. $E(T) = 5$ trials. No: every trial still has success probability 0.2, independently of previous failures.

Problem · Investigate, justify, conclude. Roll a fair die until the first 6, but stop after 10 rolls in any case. Let N be the number of rolls actually made. Find $P(N = 10)$, the probability of obtaining at least one 6, and E(N). For the mean, use $E(N) = \sum P(N > n)$, from $n = 0$ to 9.

Solution. $N = 10$ when the first nine rolls fail, regardless of the tenth: $P(N = 10) = (\frac{5}{6})^9$. The probability of at least one 6 is $1 - (\frac{5}{6})^{10}$. For $0 \leq n \leq 9$, $P(N > n) = (\frac{5}{6})^n$, giving $E(N) = 6[1 - (\frac{5}{6})^{10}]$, about 5.031 rolls. Truncation reduces the expectation below 6.